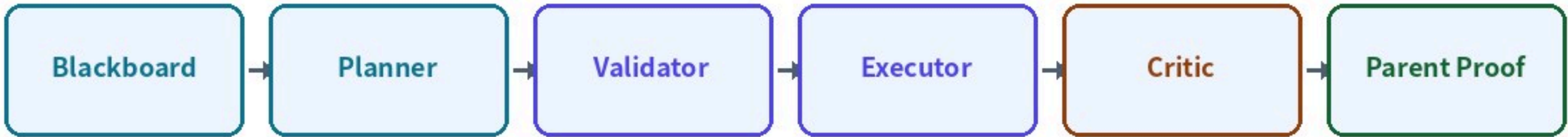


Agentic Blackboard Case Report

新架构 Agent 决策过程展示

Target	MLA-like alkaloid case
Heavy atoms / rings	25 / 5
Final verdict	fake_closed_rejected (fake_closed_rejected)
Test gate	pytest -q: 279 passed, 2 skipped, 7 warnings in 76.17s



Core message

The controller is no longer a fixed failure -> scout -> extract -> rerun chain. Each round reads the blackboard, selects typed actions, validates them, executes only whitelisted tools, and refuses solved unless deterministic parent proof exists.

Architecture Decisions

Policy-driven DAG + blackboard, with deterministic gates

1. Planner selects typed actions only; it cannot emit route SMILES or solved verdicts.
2. Action batches are schema-validated, budgeted, and stale-action checked before execution.
3. Failure critic turns route-verifier and plugin-runtime failures into bridge tasks.
4. Exact literature rows and analogy are separated; analogy only changes priorities.
5. Final solved requires stitched_parent_route_proof.v1, not backend or child solved flags.

Blackboard state sources

route_failures: 4 | bridge_tasks: 10 | terminal_blacklist: 1 | exact_rows: 1 |
selected_analogies: 3

Round-by-Round Agent Decisions

Each action has rationale, expected artifact, and success condition

Round 1 - generate_disconnection_hypotheses [accepted]

Rationale: Initial MLA-like target needs target-side handles before any rerun.

Artifact: target_side_disconnection_hypotheses.v1

Success: Aryl ester, imide, cage, and amine advisory tasks appear.

Round 1 - build_failure_critic_report [accepted]

Rationale: Prior route verifier shows large atom jump and advanced terminal; normalize that into bridge tasks.

Artifact: failure_critic_report.v1

Success: Target bridge, terminal blacklist, and next-action bias are recorded.

Round 1 - search_literature [accepted]

Rationale: Bridge tasks need target-proximal source candidates before exact replay.

Artifact: literature_scout_report.v1

Success: Scout emits source candidates and extraction recommendations.

Round 2 - compile_exact_literature_rows [accepted]

Rationale: Mock source-detail row is available; compile it as exact evidence, not as proof.

Artifact: compiled exact literature rows

Success: One exact row enters literature_evidence.exact_rows.

Round 2 - rank_analogical_hypotheses [accepted]

Rationale: Rank advisory hypotheses after exact-row context is present.

Artifact: analogical_hypothesis_ranking.v1

Success: Selected hypotheses carry required verification and no_solved_claim.

Round 2 - stop_unresolved [accepted]

Rationale: No stitched parent proof exists; stop without solved claim.

Artifact: unresolved stop marker

Success: Final verdict stays unresolved or partial, never solved.

Blackboard Updates

Failure evidence becomes typed bridge tasks instead of final-report-only text

Route failures

- large_atom_jump
- literature_template_plugin_not_invoked
- advanced_same_scaffold_terminal
- plugin_product_hits=0

Bridge tasks

- target_proximal_bridge_required: Find a target-proximal intermediate that explains the large skeleton change.
- bridge_to_literature_product_required: Find a target-side bridge before replaying exact source products.
- upstream_terminal_synthesis: Find upstream synthesis for rejected same-scaffold terminal.
- target_side_bridge_before_source_replay: Treat literature rows as disconnected until a target-proximal bridge is found.
- target_proximal_bridge: core alcohol or phenol plus activated anthranilate/imide acid fragment
- target_proximal_bridge: preassembled imide or succinimide-containing acid/activated fragment
- target_proximal_bridge: target-proximal cage intermediate with matched ring system

Terminal blacklist

- CN1CC2CCC1CC2OC(C)=O

Selected advisory hypotheses

- target_side_analogue_diene_iminium_scaffold_inspiration score=15
- target_side_analogue_diene_iminium_scaffold_inspiration score=15
- target_side_aryl_ester_or_anthranilate_disconnection score=15

Verification And Final Gate

Tests passed; solved still requires parent-route proof

Comprehensive test gate: pytest -q: 279 passed, 2 skipped, 7 warnings in 76.17s

Final verdict is fake_closed_rejected with route_status fake_closed_rejected. This is intentional: exact rows and analogical ranking are evidence for exploration, not proof. A solved claim would require stitched_parent_route_proof.v1 with target equivalence, parent verifier acceptance, stock audit, no unexplained large atom jump, child-parent connectivity, and exact-literature connectivity.

Key artifact refs

- case_r1_build_failure_critic_report_failure_critic_report_v1: /root/autodl-tmp/AutoPlanner/docs/agentic_blackboard/report_20260609/case_run/case_r1_build_failure_critic_report_failure_critic_report_v1.json
- case_r1_generate_disconnection_hypotheses_target_side_disconnection_hypotheses_v1: /root/autodl-tmp/AutoPlanner/docs/agentic_blackboard/report_20260609/case_run/case_r1_generate_disconnection_hypotheses_target_side_disconnection_hypotheses_v1.json
- case_r1_search_literature_literature_scout_report_v1: /root/autodl-tmp/AutoPlanner/docs/agentic_blackboard/report_20260609/case_run/case_r1_search_literature_literature_scout_report_v1.json
- case_r2_compile_exact_literature_rows_compiled_exact_literature_rows_mock_v1: /root/autodl-tmp/AutoPlanner/docs/agentic_blackboard/report_20260609/case_run/case_r2_compile_exact_literature_rows_compiled_exact_literature_rows_mock_v1.json
- case_r2_rank_analogical_hypotheses_analogical_hypothesis_ranking_v1: /root/autodl-tmp/AutoPlanner/docs/agentic_blackboard/report_20260609/case_run/case_r2_rank_analogical_hypotheses_analogical_hypothesis_ranking_v1.json
- case_r2_stop_unresolved_agent_stop_unresolved_v1: /root/autodl-tmp/AutoPlanner/docs/agentic_blackboard/report_20260609/case_run/case_r2_stop_unresolved_agent_stop_unresolved_v1.json
- r1_auto_failure_critic_failure_critic_report_v1: /root/autodl-tmp/AutoPlanner/docs/agentic_blackboard/report_20260609/case_run/r1_auto_failure_critic_failure_critic_report_v1.json